

Roll No.

TMA-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, MAY, 2022

ENGINEERING MATHEMATICS—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Solve the differential question :

(CO1)

$$\left(\frac{y}{x} \sec y - \tan y \right) dx + (\sec y \log x - x) dy = 0$$

OR

(b) Define exact equation. Solve :

(CO1)

$$(x^4 y^4 + x^2 y^2 + xy) y dx + (x^4 y^4 - x^2 y^2 + xy) x dy = 0$$

2. (a) Solve :

(CO1)

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 8x^2 e^{2x} \cos 2x.$$

P. T. O.

OR

(b) Define linear differential equation with constant coefficient. Solve :

(CO1)

$$\frac{d^2 y}{dx^2} + a^2 y = \tan ax$$

3. (a) Solve :

(CO1)

$$(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \{ \log(1+x) \}$$

OR

(b) Define Laplace transform and solve :

(CO2)

$$\int_0^{\infty} e^{-pt} t^3 \cos t \, dt.$$

4. (a) Solve by variation of parameter :

(CO1)

$$x^2 y'' + xy' - y = x^2 e^x$$

OR

(b) Solve :

(CO2)

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + y = t$$

given that $y = -3$ when $t = 0$, $y = -1$, when $t = 1$.

5. (a) Define Theorem. Use Convolution theorem to find $L^{-1} \left\{ \frac{P}{(p^2 + 4)^2} \right\}$.

(CO2)

(3)

OR

- (b) (i) Define the condition of function of exponential order and prove that $F(t) = t^2$ is of exponential order 3. (CO2)

(ii) Find :

$$L^{-1} \left\{ \frac{p^3 - 3p^2 + 6p - 4}{(p^2 - 2p + 2)^2} \right\}$$

TEE-201

Roll No.

B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, May, 2022

BASIC ELECTRICAL ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) (i) Discuss basic characteristics of inductor, resistor and capacitor.
(ii) In the circuit of Fig. 1, find i_s , given $i_L = e^{-4t}$: 10 Marks (CO1)



Fig. 1

OR

- (b) Prove the maximum power transfer theorem. Also, derive the expression for maximum power and efficiency. 10 Marks (CO1)

P. T. O.

2. (a) For the circuit shown in Fig. 2, find the Thevenin equivalent as viewed by the resistance R . Find the value of R for maximum power dissipation in it and the value of this power : 10 Marks (CO2)

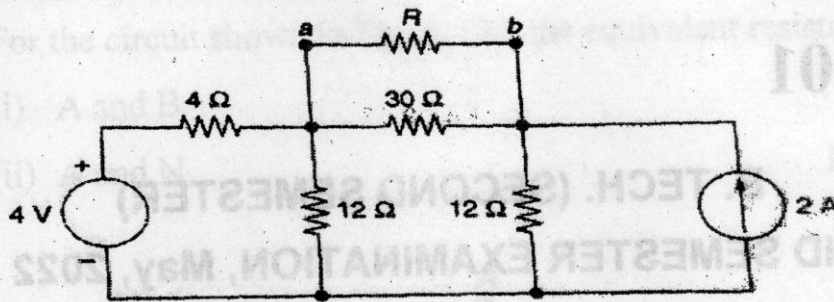


Fig. 2

OR

- (b) In the circuit of Fig. 3 : 10 Marks (CO2)

(i) $V_s = 16\text{ V}$ and (ii) $I_s = 10\text{ A}$, find the value of I .

Using the principle of superposition theorem.

Also write the statement of superposition theorem :

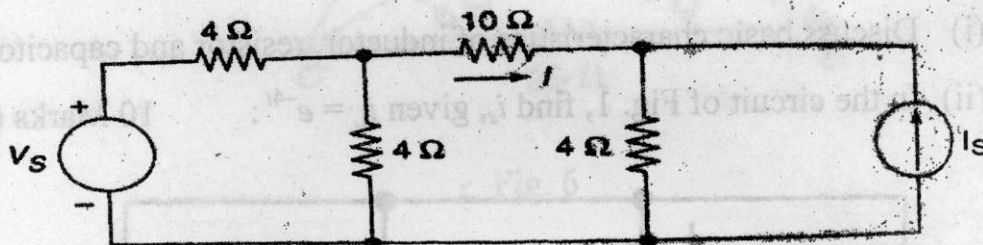


Fig. 3

3. (a) Find the average value, RMS value, Peak factor and form factor of the full wave rectifier sine wave shown in Fig. 4 : 10 Marks (CO3)

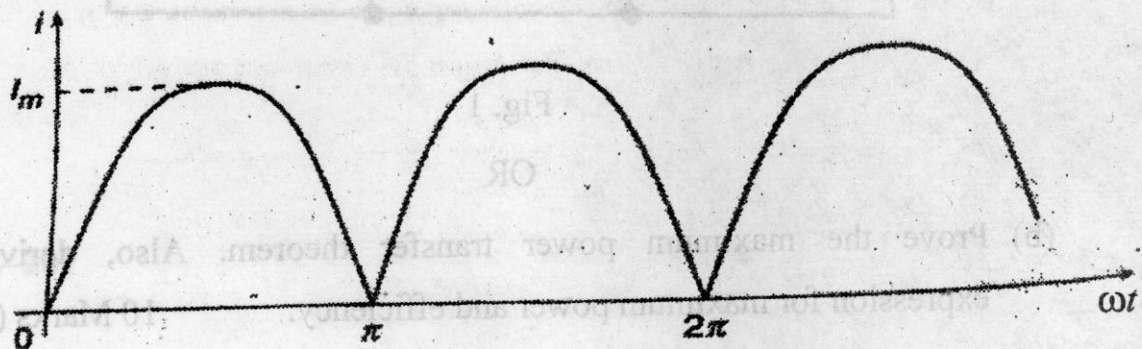


Fig. 4

OR

- (b) A coil of resistance 8 ohm and inductance 0.1 H is connected in series with a condenser of capacitance 160 μF across 230 V, 50 Hz supply.

Calculate :

10 Marks (CO3)

- (i) The inductive reactance
- (ii) The capacitive reactance
- (iii) The circuit impedance
- (iv) The circuit current
- (v) The circuit power factor
- (vi) The coil and condenser voltages respectively

4. (a) By using the phasor diagram, find the resultant (magnitude) and phase angle :

10 Marks (CO3)

$$\bar{V} = 10 \angle 0^\circ + 10 \angle 60^\circ - 10\sqrt{3} \angle 30^\circ$$

OR

- (b) For the circuit shown in Fig. 5, find the nodal voltages and the current through the 2 ohm resistance :

10 Marks (CO2)

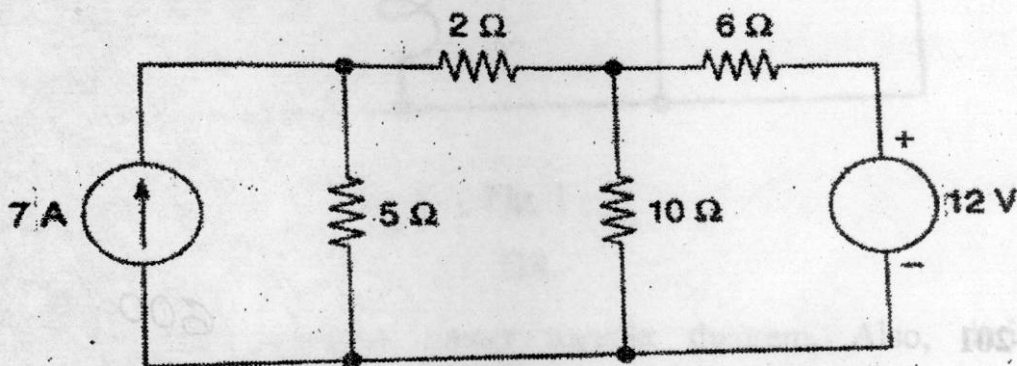


Fig. 5

P. T. O.

5. (a) Calculate the average power for series RL circuit. 10 Marks (CO3)

OR

- (b) For the circuit shown in Fig. 6, find the equivalent resistance between :

(i) A and B

(ii) A and N

10 Marks (CO2)

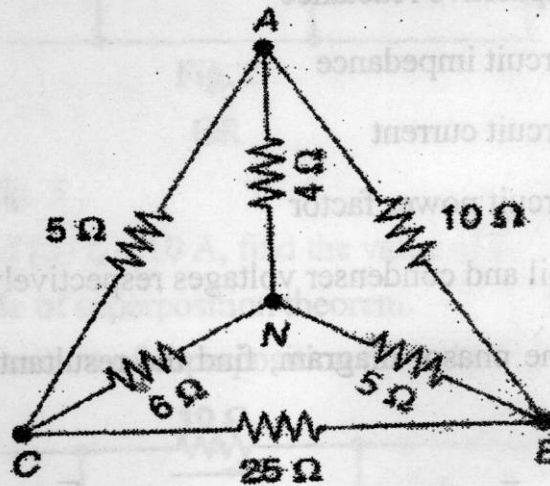


Fig. 6

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BTE-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, May, 2022**

ADVANCED ORGANIC CHEMISTRY

Time : $1\frac{1}{2}$ Hours

Maximum Marks : 50

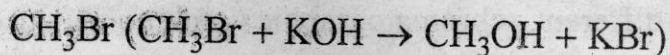
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What is a carbocation and a carbanion ? Explain the ways to determine the stability of carbocation and carbanion. (CO2)

OR

- (b) Write a short note on mesomeric effect and electromeric effect. (CO2)

2. (a) S_N1 reaction proceeds with the formation of racemized products. Explain with the help of hydrolysis of : (CO2)



OR

- (b) Differentiate between S_N1 and S_N2 reaction mechanism by taking the example of hydrolysis of CH_3Br . (CO2)

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(2)

3. Explain why benzene undergoes electrophilic substitution reactions whereas it undergoes addition reactions. Classify the following groups as activating or deactivating with respect to further electrophilic substitution of the aromatic ring : (CO2)

(i) $-\text{NH}_2$

(ii) $-\text{NO}_2$

(iii) $-\text{SO}_3\text{H}$

(iv) $-\text{Cl}$

(v) $-\text{CN}$

(vi) $-\text{CH}_3$

4. (a) What are nanomaterials ? Differentiate between nanomaterials and bulk materials. (CO3)

OR

- (b) What is Green Chemistry ? Elaborate on the 12 principles of green chemistry. (CO3)

5. (a) What are the different separation methods used in purification of organic compounds ? Explain the different distillation processes used for separation and purification. (CO1)

OR

- (b) Classify the different chromatographic techniques used. Explain the mechanism of separation between adsorption and partition chromatography.

(CO1)

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THU-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, May, 2022

ADVANCED PROFESSIONAL COMMUNICATION

Time : $1\frac{1}{2}$ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any **one** of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. Write an essay in 300 words on the topic :

(a) My favorite childhood memory

(CO1)

OR

(b) Do social media violate our privacy ?

(CO1)

2. Develop a paragraph of 150 words on the topic :

(a) A dark hallway

(CO1)

OR

(b) I like rainy season because

(CO1)

3. Compose a précis on the following :

(a) There are hundreds of superstitions which survive in various parts of the country, and the story of them is rather amusing. We are told, for

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example, that it is unlucky to point to the new moon or to look at it through glass, but if we bow nine times to it we shall have a lucky month.

Now suppose you tell a scientist that you believe a certain superstition—let us say, that the howling of a dog is a sign of death. The scientist will immediately require evidence before he can accept your belief. He will want figures to prove it. It will be useless to quote two or three cases; he will want hundreds. He will want also to know (i) if it ever happens that the howling of dogs is not followed by a death, (ii) if ever a person's death is predicted by the howling of dogs. The answer to the former question is in the affirmative (positive), and to the latter in the negative. Your superstition will not bear investigation. It may impress an ignorant person; but it cannot face the light of facts. Your case would not carry conviction in a Court of Law.

Apart from this process of testing by results, any intelligent man will want to know the 'reason why'. What connection can there be between a howling dog and an approaching death? Can it be cause and effect? Can it be that the dog has a gift of foreseeing such events? Or is the dog the instrument employed by some uncanny power that moves invisibly in our midst? (238 Words). (CO1)

OR

- (b) It is very easy to acquire bad habits, such as eating too many sweets or to much food, or drinking too much fluid of any kind, or smoking. The more we do a thing, the more we tend to like doing it : and, if we do not continue to do it, we feel unhappy. This is called the force of habit, and the force of habit should be fought against.

Things which may be very good when only done from time to time, tend to become very harmful done too often and too much. This applies even to such good things as work or rest. Some people form a bad habit of working too much, and others of idling too much. The Wiseman always remembers that this true about himself and checks any bad habit. He says to himself. "I am now becoming idle", or "I like too many sweets", or "I smoke too much" and then adds, "I will get myself out of this bad habit at once."

One of the most widely spread of bad habits is the use of tobacco. Tobacco is now smoked or chewed by men, often by women, and even by children, almost all over the world. It was of Europe from America by Sir Walter Rayleigh four centuries ago, and has thence spread everywhere. I very much doubt whether there is any good in the habit, even when tobacco is not used in excess; and it is extremely difficult to get rid of the habit when once it has been formed.

Alcohol is taken in almost all cool and cold climates, and to a very much less extent in hot ones. Alcohol is not necessary in any way to anybody. Millions of people are beginning to do without it entirely; and once the United States of America have passed laws which forbid its manufacture or sale throughout the length and breadth of their vast country. The regular use of alcohol, even in small quantities, tends to cause mischief in many ways to various organs of the body. It affects the liver; it weakens the mental powers, and lessens the general energy of the body. (234 words)

(CO1)

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4. Attempt as per the directions against each part :

(CO2)

(a) (i) Ravya and Ridhan are *fair weather friends* as they always support each other in need. (Correct the usage of italicized idiom)

(ii) The detectives discovered that the documents that were important to the investigation were *no more* in *their cupboard*.

(Provide a suitable oxymoron for the italicized phrase)

(iii) Samaira was *raised and taken care of* by her aunt after her parents kicked the bucket.

(Provide a suitable phrasal verb for the italicized phrase)

(iv) He *worked hard* to clear the National Eligibility-Cum-Entrance Test. (Replace the italicized phrase with an appropriate idiom)

(v) The dress wore by Rihanna at Met Gala was *terribly good*. (Replace the italicized oxymoron with one meaning the opposite)

OR

(b) (i) The *sudden eruption* of a silent Volcano took lives of a lot of tourists. (Replace the italicized phrase with a suitable phrasal verb)

(ii) Tom and Jerry should now *act mature and stop fighting* the way they do. (Replace the italicized phrase with an appropriate idiom)

(iii) Mr. Joe Clinton was welcomed with a *low level of noise* by the US senate. (Provide a suitable oxymoron for the italicized phrase)

(iv) If a celebrity feels that he can *manage* the stardom without a dynamic PR team, he is sadly mistaken.

(Provide a suitable phrasal verb for the italicized word without altering the meaning of the sentence)

- (v) All the students are *confused* as to how to answer the questions 4 and 5 of this examination. (Replace the italicized word with an appropriate idiom meaning the same)

5 Attempt as per the directions against each part :

- (a) (i) The urge to earn a lot of money, workload and stress has turned the generation of today into a _____.

(Fill in the blank using an appropriate oxymoron)

- (ii) The TV actors were seen _____ with each other on the eve of the award function.

(Fill in the blank using an appropriate phrasal verb)

- (iii) The journalists have _____ about the ethics of journalism.

(Fill in the blank with an idiom meaning started to understand)

- (iv) Stand-up comedian and actor Bharti Sharma did not _____ performing on the stage even during her pregnancy.

(Fill in the blank using an appropriate oxymoron)

- (v) The release of the film was _____ because the censor board did not give the film a green signal.

(Fill in the blank using an appropriate phrasal verb)

OR

- (b) (i) The teacher was feeling _____ therefore she did not teach during the class. (Fill in the blank using an appropriate idiom meaning ill.)

- (ii) My trip to Singapore turned into _____ when upon reaching there I received an e-mail stating that I had to submit my final year project in 7 days. (Fill in the blank using an appropriate oxymoron)

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(iii) The actor had to _____ his tears at his mother's funeral because of the presence of his daughter who loved her grandmother terribly.

(Fill in the blank using an appropriate phrasal verb)

(iv) The movie that released last month was a _____ because the new movie that got released this Friday is a superhit.

(Fill in the blank using an appropriate idiom)

(v) Katrina marrying Vicky was an _____ as fans were aware of it even before the news came out into the media.

(Fill in the blank using an appropriate oxymoron)

OR

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TMA-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, MAY, 2022

ENGINEERING MATHEMATICS—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Solve the differential question :

(CO1)

$$\left(\frac{y}{x} \sec y - \tan y \right) dx + (\sec y \log x - x) dy = 0$$

OR

(b) Define exact equation. Solve :

(CO1)

$$(x^4 y^4 + x^2 y^2 + xy) y dx + (x^4 y^4 - x^2 y^2 + xy) x dy = 0$$

2. (a) Solve :

(CO1)

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 8x^2 e^{2x} \cos 2x.$$

P. T. O.

OR

(b) Define linear differential equation with constant coefficient. Solve :

(CO1)

$$\frac{d^2 y}{dx^2} + a^2 y = \tan ax$$

3. (a) Solve :

(CO1)

$$(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \{ \log(1+x) \}$$

OR

(b) Define Laplace transform and solve :

(CO2)

$$\int_0^{\infty} e^{-pt} t^3 \cos t \, dt.$$

4. (a) Solve by variation of parameter :

(CO1)

$$x^2 y'' + xy' - y = x^2 e^x$$

OR

(b) Solve :

(CO2)

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + y = t$$

given that $y = -3$ when $t = 0$, $y = -1$, when $t = 1$.5. (a) Define Theorem. Use Convolution theorem to find $L^{-1} \left\{ \frac{P}{(p^2 + 4)^2} \right\}$.

(CO2)

(3)

OR

- (b) (i) Define the condition of function of exponential order and prove that $F(t) = t^2$ is of exponential order 3. (CO2)

(ii) Find :

$$L^{-1} \left\{ \frac{p^3 - 3p^2 + 6p - 4}{(p^2 - 2p + 2)^2} \right\}$$

Roll No.

TEC-201

B. TECH. (SECOND SEMESTER) MID SEMESTER EXAMINATION, MAY, 2022

BASIC ELECTRONICS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any **one** of the sub-questions.
(ii) Each sub-question carries 10 marks.

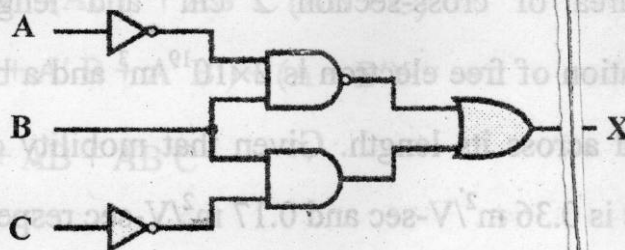
1. (a) Subtract $(7476.23)_9 - (783.5)_9$ by r 's complement method. (CO1)
Subtract $(FBD9.E3)_{16} - (A2C.A)_{16}$ by $(r-1)$'s complement method.

OR

- (b) Minimize the following by K-map and realize the minimized function by NAND and NOR gate only : (CO1)

$$F(A, B, C, D) = \prod M(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$$

2. (a) Derive Boolean expression for output X for given digital circuit and sketch truth table : (CO1)



P. T. O.

OR

(b) (i) $(7653.21)_8 (?)_2 = (?)_{16}$

(ii) $(56543)_7 + (53214)_7 - (3264)_8 = (?)_7$

(iii) If $(211)_x = (152)_8$, find the value of base x . (CO1)

3. (a) (i) Explain formation of energy bands in a solid crystal. Differentiate conductor, semiconductor and insulator based on energy band diagram. (CO2)

- (ii) In a doped semiconductor, there are 4.52×10^{24} holes/m³ and 1.25×10^{14} electrons/m³. What will be the carrier density of undoped specimen?

If electron and hole mobilities are $0.38 \text{ m}^2/\text{V-sec}$ and $0.18 \text{ m}^2/\text{V-sec}$ respectively, determine conductivity of intrinsic and doped semiconductor.

OR

- (b) (i) Briefly explain drift current and diffusion current in semiconductors. (CO2)

- (ii) Calculate the drift current produced in an intrinsic Ge semiconductor having area of cross-section 2 cm^2 and length 0.4 mm , if concentration of free electron is $2 \times 10^{19}/\text{m}^3$ and a battery of 2 volts is applied across its length. Given that mobility of free electrons and holes is $0.36 \text{ m}^2/\text{V-sec}$ and $0.17 \text{ m}^2/\text{V-sec}$ respectively.

(3)

4. (a) (i) Define mobility and drift velocity. Prove that conductivity of a conductor is given by $\sigma' = \mu n e$. (CO2)
- (ii) A small concentration of minority carries is injected into a homogeneous semiconductor crystal at one point. An electric field of 10 V/cm is applied across the crystal and this moves the minority carries a distance of 1 cm in 20 μ -sec. Determine mobility (in $\text{cm}^2/\text{V-sec}$).

OR

- (b) (i) If N_D and N_A are donor and acceptor impurities and n_i is the intrinsic concentration, establish the relation for minority and majority charge densities in extrinsic semiconductors. (CO2)
- (ii) A bar of a pure silicon has cross-sectional area of 1 mm^2 and intrinsic concentration of silicon is $1.5 \times 10^{16} \text{ m}^{-3}$. The free electron and hole mobilities are $0.13 \text{ m}^2/\text{V-sec}$ and $0.05 \text{ m}^2/\text{V-sec}$ respectively. Find the conductivity and length of bar whose resistance is $50 \text{ k}\Omega$.
5. (a) Explain the formation of barrier potential in PN junction diode along with its I-V characteristics. (CO1 & CO2)

OR

- (b) Simplify the following expressions with the help of Boolean algebra :

(CO1 & CO2)

$$Y = AB + (AC)' + AB' C (AB + C)$$

$$Y = (A + A' B + A' B') + (A + B)'$$

$$Y = A + AB + AB' C$$

Roll No.

TPH-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, May, 2022

ENGINEERING PHYSICS

Time : $1\frac{1}{2}$ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Using Fresnel biprism experiment, discuss the method for determining various parameters required for calculation of wavelength of monochromatic light. (CO1)

OR

- (b) Derive the expression for path difference in a wedge shape film in reflected light. Also write the condition for maxima and minima. (CO1)
2. (a) Describe Newton's ring method for measuring the refractive index of a transparent liquid. Give the necessary theory. (CO1)

OR

- (b) In Newton's ring are observed in reflected light of wavelength $5500 \text{ \AA}$. The diameter of the 10th dark ring is 0.50 cm . Find the radius of curvature of the lens and the thickness of the film. (CO1)

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(2)

3. (a) In N-slit Fraunhofer diffraction using the expression for resultant intensity, derive the intensity of principal maxima and secondary maxima. Also discuss the effect of increasing or decreasing the number of slits on the intensity pattern. (CO1)

OR

- (b) Discuss Rayleigh criteria of least resolution. Derive the expression for resolving power of a grating. If a plane transmission grating has 15000 lines per inch. Calculate grating element and the resolving power in second order. (CO1)
4. (a) Derive the expression for numerical aperture and acceptance angle in an optical fiber. (CO2)

OR

- (b) Discuss the working principle of fiber optics. Also distinguish between various types of optical fiber. (CO2)
5. (a) An optical fiber the refractive index of the core and cladding are 1.42 and 1.40 respectively. Calculate the numerical aperture, acceptance angle and critical angle. (CO2)

OR

- (b) In an optical fiber, explain the following terms :
numerical aperture, acceptance angle, acceptance angle of cone, fractional refractive index change.
Also write applications of optical fibers. (CO2)

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TCH-201

B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, May, 2022
ENGINEERING CHEMISTRY

Time : $1\frac{1}{2}$ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Write the magnetic properties of N_2 and O_2 molecules with the help of MOT. (CO1)

OR

- (b) With the help of band theory, explain conductors, semiconductors and insulators. (CO1)

2. (a) What are the consequences of hydrogen bonding ? Explain inter and intramolecular hydrogen bonding with suitable examples. (CO1)

OR

- (b) What do you mean by regeneration of zeolite softener ? 10000 litres of hard water was softened by zeolite process. The zeolite was regenerated by passing 200 litres of sodium chloride solution containing 1500 mg/L of NaCl. Calculate hardness of water. (CO2)

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(2)

3. (a) Name the impurities, causing temporary and permanent hardness in water. Explain the principle with the help of reactions, of the water softening process in which the soluble impurities are precipitated as sludge. Also give *two* disadvantages of the process. (CO2)

OR

- (b) Calculate the amount of lime and soda required for softening of 5,000 litres of water containing the following salts :

$\text{Mg}(\text{HCO}_3)_2 = 13 \text{ ppm}$, $\text{MgSO}_4 = 18 \text{ ppm}$, $\text{Ca}(\text{HCO}_3)_2 = 25 \text{ ppm}$,
 $\text{MgCl}_2 = 35 \text{ ppm}$ and $\text{NaCl} = 55 \text{ ppm}$.

4. (a) Write short notes on the following : (CO1)
- (i) Electronic transition in UV-Visible spectroscopy
 - (ii) Calgon conditioning

OR

- (b) (i) Classify polymers on the basis of their tacticity. (CO3)
- (ii) Thermoplastic and Thermosetting Polymers.

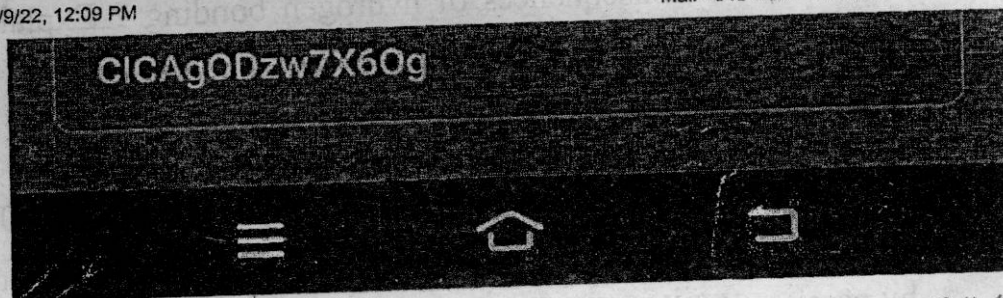
5. (a) What do you mean by conducting polymers ? Discuss various classes of conducting polymers. (CO3)

OR

- (b) Write the preparation, properties, and applications of Dacron and Bakelite. (CO3)

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TCH-201

Roll No.

MAB-201

B. TECH. (BIOTECHNOLOGY) (SECOND SEMESTER)

MID SEMESTER EXAMINATION, MAY, 2022

ENGINEERING MATHEMATICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Solve the following system of equations by using matrix method :

$$x + y = 0$$

$$y + z = 1$$

$$x + z = 3$$

OR

(b) Find the inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$. (CO1)

P. T. O.

2. (a) Define singular and non-singular matrix with example and if

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}, \text{ then find :} \quad (\text{CO1})$$

(i) AA^T

(ii) $A^T A$.

OR

(b) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$.

(CO1)

3. (a) Show that the matrix $A = \begin{bmatrix} -8 & 5 \\ 2 & 4 \end{bmatrix}$ satisfies the matrix equation

$$A^2 + 4A - 42I = O, \text{ where } I \text{ is identity matrix and } O \text{ is null matrix.}$$

(CO1)

OR

- (b) Check the continuity of the function :

(CO1)

$$f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} \leq x < 0 \\ 3 - 2x, & 0 \leq x \leq \frac{3}{2} \\ -3 - 2x, & x \geq \frac{3}{2} \end{cases}$$

4. (a) Find the maximum and minimum values of the function : (CO2)

$$y = x^3 - 3x^2 - 9x + 27$$

(3)

OR

(b) Evaluate the following :

(CO2)

(i) $\int x^4 \log x \, dx$

(ii) $\int x\sqrt{2x+1} \, dx$

5. (a) Find $\frac{dy}{dx}$ of the following with respect to x :

(CO2)

(i) $y = \frac{1+x^2}{1-x^2}$

(ii) $y = \frac{1+e^x}{1-e^x}$

OR

(b) Evaluate the following :

(CO2)

(i) $\lim_{x \rightarrow 0} \frac{x}{\sqrt{2+x} - \sqrt{2-x}}$

(ii) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}$

OR

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